```
#include <iostream>
using namespace std;

const int MAX = 5;

class Stack
{
private:
   int a[MAX];
   int top;

public:
   Stack()
   {
      top = -1;
   }

   void push(int x)
   {
      top++;
      a[top] = x;
   }

   int pop(void)
   {
      int x = a[top];
      top--;
      return x;
   }

   int peek(void)
   {
      return a[top];
   }

   int full(void)
   {
      if (top == MAX - 1)
      {
         return 1;
      }
      else
      {
         return 0;
      }
   }

   int empty(void)
```

```
    {
        if (top == -1)
        {
            return 1;
        }
        else
        {
            return 0;
        }
    }

    void display(void)
    {
        for (int i = top; i >= 0; i--)
        {
            cout << endl << a[i];
        }
    }
};

int main()
{
    int x;
    Stack s;

    if (!s.full()) s.push(44);
    if (!s.full()) s.push(78);
    if (!s.full()) s.push(56);
    if (!s.full()) s.push(27);
    if (!s.full()) s.push(100);
    if (!s.full()) s.push(45);

    cout << "Stack elements:";
    s.display();

    if (!s.empty())
    {
        x = s.pop();
        cout << endl << "Removed: " << x;
    }

    cout << endl << "Stack after pop:";
    s.display();

    if (!s.empty())
    {
        x = s.peek();
        cout << endl << "Top element is: " << x;
```

```
    }

    return 0;
}
//[Running] cd "c:\Users\STUDENT\Desktop\manasi kulkarni 209\" && g++
stack_usingarray.cpp -o stack_usingarray && "c:\Users\STUDENT\Desktop\manasi kulkarni
209\"stack_usingarray
//Stack elements:
//100
//27
//56
//78
//44
//Removed: 100
//Stack after pop:
//27
//56
//78
//44
//Top element is: 27
//[Done] exited with code=0 in 0.368 seconds
```